higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 3

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This marking guideline consists of 5 pages.

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QUESTION 1

1.1
- Aggregate
- Filler
- Bitumen (3)

1.2
- To improve the riding comfort for the road users
- To ensure that the required level thickness and surface grading are met
- To even out any bad undulation in the base of the road (3)

1.3
- As a drainage
- Agricultural underpass
- A pedestrian crossing
- Other possible purposes (4)

1.4 1.4.1 This is the part of the road between shoulder break✓ point that traffic drive over.✓

1.4.2 This is a profile of what the level✓ of the road will look like when it is completed.✓

1.4.3 It is the most direct✓ and practical✓ route to move material.

1.4.4 It is the part of the road between the OGL✓ and the outer boundaries of the slope of cutting-fill and side drains.✓

1.4.5 It is the material taken from a designated borrow area✓ or from dump used to build roads.✓

(5 × 2) (10)
[20]

QUESTION 2

2.1
- The reduction in noise pollution
- To improve safety
- Comfort and durability (3)

2.2
- Layer must be swept and cleaned using rotary broom to remove loose material.
- Material swept to the side of the road and any areas that the rotary broom cannot reach must be swept using hand broom.
- All sweeping must be done carefully so that the layer will not be damaged.
- After it has been swept, the surface will be inspected to check that it meets the specifications.
- Before the prime is applied, the base is sprayed lightly with water. (5)

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2.3
- Gap-graded asphalt
- Open-graded asphalt
- Continuously graded asphalt (3)

2.4 Because it will spread the asphalt at the correct thickness and width (1)

2.5
- Aggregate
- Binder
- Tack coat (3)

[15]

QUESTION 3

3.1
- Warning sign
- Regulatory sign
- Guidance and information sign (3)

3.2
- Where the embankment is higher than 4 m on surface road and higher than 6 m on dust road.
- Where a wall higher than 1 m is built next to the road and is not designed as a barrier.
- At any curve with a radius of 300 mm or less and NO recovery area.
- In the median where the median slope is more than 1:4.
- At the island of the road that divides the lanes. (5)

3.3
- Post bolt
- Spencer block
- Guardrail (2)

[10]

QUESTION 4

4.1 A horizontal curve is a circular or arched curve used wherever there is a change in the direction✓ of a road whilst a vertical curve is a parabolic curve used wherever there is a change in the slope or gradient✓ of the road. (2)

4.2
- Simple curve: this is an arch with a constant curve radius between two straight lines of a road.
- Compound curve: it is a combination of two or more simple curves that curve in the same direction.
- Reverse curve: it is a combination of two simple curves in opposite directions.
- Spiral/Transition curve: the combination of a simple and compound curve.

(Any 3 × 2) (6)

4.3
- Stopping sight distance
- Meeting sight distance
- Passing sight distance (3)

4.4
- Centering the theodolite
- Levelling the theodolite
- Removing parallax (3)

4.5
- Set up the theodolite such that it is levelled and centered.
- Point the telescope towards the object to be sighted.
- Look through the telescope and focus on the target.
- Look into the reading microscope.
- Adjust the mirror until enough light is reflected into the instrument to take clear and accurate reading.
- Twist the eye piece to make the focus clear. (6)

4.6
- Mirror
- Optical micro-meter screw
- Upper plate clamp
- Movable head
- Trivet stage
- Levelling screw (5)

[25]

QUESTION 5

5.1
- Grid roller
- Smooth drum roller
- Pneumatic roller
- Vibratory roller
- Pad-foot roller
- Impact roller
- Static roller (Any 5 × 1) (5)

5.2 5.2.1
- Taping roller
- High water content needed for compaction

5.2.2
- Taping vibrator
- These offer various strengths depending on the type of grade

5.2.3
- Vibrating smooth drum
- Compacts well when it is saturated with water and rolled with a static roller to bring fines to the surface

(3 × 2) (6)

5.3 5.3.1 Strong when it's dry and takes a long time for water to saturate it.

5.3.2 Poor cohesive strength when it's dry – it means this soil sample will crumble easily.

(2 × 1) (2)

5.4
- 152,4 mm diameter mould
- Base plate
- 25,4 mm spencer block
- 50,8 mm diameter tamper
- 300 mm long straight edge (5)

5.5 Proctor test (1)

5.6
- It allows you to determine the maximum dry density that is achievable for the material being tested.
- To determine the optimum moisture content of the soil sample. (2)

5.7 5.7.1 MC – Moisture Content

5.7.2 OMC – Optimum Moisture Content

5.7.3 RD – Relative Density

5.7.4 AASHTO – American Association of State Highway and Transportation Officials

5.7.5 CBR – California Bearing Ration

(5 × 1) (5)

5.8 The relationship is that they are all used to determine by establishing the moisture density✓ relationship of the material✓ when prepared and compacted✓ with a vibratory at moisture content.✓ (4)

[30]

TOTAL: 100

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